

3 实验三 数据操作实验

3.1 实验目的

- ①了解和掌握 Sharedpreferences 的使用；
- ②了解和掌握普通文件的读写操作；
- ③了解和掌握 SQLiteDatabase 的创建、编辑和查询功能。

3.2 实验内容

利用已学的知识，综合利用 Sharedpreferences、file、contentprovider、Sqlite，完成一个用户注册与登录程序，同时支持用户信息保存的功能，设计 1~2 个简单表格，完成数据的插入、删除、修改和查询等基本数据库操作。

创建表格 student(no,name,provinceID)和表格 province(ID,name)

完成如下操作：

- 1、省份信息的增、删改和查询工作
- 2、学生信息的增、删改和查询工作
- 3、为了减少冗余信息的存储，学生表格中仅存储省份的 ID 信息，但操作时显示省份名称信息。
- 4、数据库文件存放在 SD 卡中。
- 5、登录信息（用户名，密码）需要存放在 Sharedpreferences 中。

3.3 实验体会及评语

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实验体会			
本次实验中，我深入学习并掌握了 Android 开发中 SQLite 数据库的使用方法。通过创建 StudentDatabaseHelper 类，我了解了如何在应用中创建和管理本地数据库，以及如何在 SQLite 中定义表结构、插入初始数据。在项目开发过程中，我实现了点击列表项显示详细信息、长按删除数据等功能，巩固了对数据库基本操作（增删改查）的理解。 在删除操作中，采用了参数化删除语句，避免了 SQL 注入问题，并通过事务处理和异常捕获，提升了程序的健壮性和用户体验。同时，我学习了如何通过更新 ListView 的数据源来同步界面显示，确保数据一致性。 这次实验让我更熟悉了 Android 中的本地数据持久化技术，提升了开发实际应用的能力，为今后处理更复杂的数据存储与管理任务打下了坚实的基础。			
问题及解决办法			
问题描述：数据库无法正常链接。 解决方法：修改程序运行权限。 问题描述：addButton 按钮无法正常点击。 解决方法：在 FrameLayout 布局中，将 addButton 按钮置于顶层。 问题描述：输入框无法输入中文。 解决方法：为输入框添加一个属性：android:inputType="text"。 问题描述：用户无法正常。 解决方法：由于表结构和注册/登录逻辑不一致，需要修改 registerBTN.setOnClickListener() 里面的插入逻辑，把 "id" 改成 "STUDENTID"。			
成绩			
评语			

3.4 实验记录（内容不够可以添加）

主要布局文件

<LinearLayout

```
    android:id="@+id/login"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/app">
```

<androidx.cardview.widget.CardView

```
    android:layout_width="337dp"
    android:layout_height="365dp"
    android:layout_marginHorizontal="35dp"
    android:layout_marginVertical="150dp"
    android:background="#9FFFFFFF"
    app:cardBackgroundColor="#ABFFFFFF"
    app:cardCornerRadius="25dp">
```

<TextView

```
    android:id="@+id/textView"
    android:layout_width="337dp"
    android:layout_height="360dp" />
```

<LinearLayout

```
    android:layout_width="match_parent"
    android:layout_height="match_parent"
```

```
android:orientation="vertical">
```

```
<TextView
```

```
    android:id="@+id/welcome"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginStart="100dp"
    android:layout_marginTop="10dp"
    android:text="@string/welcome"
    android:textColor="@color/dark_brown"
    android:textSize="30sp"
    android:textStyle="bold" />
```

```
<TextView
```

```
    android:id="@+id/ID_input"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginStart="55dp"
    android:layout_marginTop="20dp"
    android:text="@string/请输入您的学号"
    android:textColor="@color/dark"
    android:textStyle="italic" />
```

```
<LinearLayout
```

```
    android:layout_width="match_parent"
    android:layout_height="50dp"
    android:layout_marginStart="10dp"
    android:layout_marginTop="5dp"
    android:orientation="horizontal">
```

```
<ImageView
    android:id="@+id/peopleID"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_margin="5dp"
    android:layout_weight="1"
    android:contentDescription="@string/todo"
    app:srcCompat="@drawable/people" />
```

```
<EditText
    android:id="@+id/ID"
    android:layout_width="261dp"
    android:layout_height="50dp"
    android:layout_marginStart="8dp"
    android:layout_marginEnd="20dp"
    android:ems="10"
    android:hint="@string/studentid"
    android:inputType="text" />
```

```
</LinearLayout>
```

```
<TextView
    android:id="@+id/PASSWORD_input"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginStart="55dp"
    android:layout_marginTop="15dp"
    android:text="@string/请输入您的密码"
    android:textColor="@color/dark"
```

```
        android:textStyle="italic" />
```

```
<LinearLayout
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="50dp"
```

```
    android:layout_marginStart="10dp"
```

```
    android:layout_marginTop="10dp"
```

```
    android:orientation="horizontal">
```

```
<ImageView
```

```
    android:id="@+id/PASSWORD"
```

```
    android:layout_width="0dp"
```

```
    android:layout_height="wrap_content"
```

```
    android:layout_margin="5.5dp"
```

```
    android:layout_weight="1"
```

```
    app:srcCompat="@drawable/password2" />
```

```
<EditText
```

```
    android:id="@+id/password"
```

```
    android:layout_width="261dp"
```

```
    android:layout_height="50dp"
```

```
    android:layout_marginStart="8dp"
```

```
    android:layout_marginEnd="20dp"
```

```
    android:ems="10"
```

```
    android:hint="@string/password"
```

```
    android:inputType="textPassword" />
```

```
</LinearLayout>
```

```
<CheckBox
```

```
android:id="@+id/checkbox"
android:layout_width="wrap_content"
android:layout_height="48dp"
android:layout_gravity="end"
android:layout_marginEnd="25dp"
android:text="@string/注册后自动登录" />
```

<LinearLayout

```
style="?android:attr/buttonBarStyle"
android:layout_width="match_parent"
android:layout_height="50dp"
android:layout_marginStart="20dp"
android:layout_marginTop="1dp"
android:layout_marginEnd="20dp"
android:orientation="horizontal">
```

<Button

```
android:id="@+id/register_btn"
style="?android:attr/buttonBarButtonStyle"
android:layout_width="20dp"
android:layout_height="wrap_content"
android:layout_marginStart="30dp"
android:layout_marginEnd="40dp"
android:layout_weight="1"
android:backgroundTint="#C97A02"
android:text="@string/注册"
android:textColor="@color/white_dark"
android:textSize="15sp" />
```

```
<Button
    android:id="@+id/login_btn"
    android:layout_width="20dp"
    android:layout_height="wrap_content"
    android:layout_marginEnd="30dp"
    android:layout_weight="1"
    android:backgroundTint="#C97A02"
    android:text="@string/登录"
    android:textColor="@color/white_dark"
    android:textSize="15sp" />
```

```
</LinearLayout>
```

```
</LinearLayout>
```

```
</androidx.cardview.widget.CardView>
```

```
</LinearLayout>
```

```
<FrameLayout
```

```
    android:id="@+id/app"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:background="#9FFFFFFF"
    app:cardBackgroundColor="#ABFFFFFF"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
```

```
app:layout_constraintTop_toTopOf="parent">
```

```
<ImageView
```

```
    android:id="@+id/imageView4"  
    android:layout_width="137dp"  
    android:layout_height="50dp"  
    android:layout_marginStart="30dp"  
    android:layout_marginTop="10dp"  
    app:srcCompat="@drawable/title1" />
```

```
<ImageView
```

```
    android:id="@+id/imageView3"  
    android:layout_width="211dp"  
    android:layout_height="40dp"  
    android:layout_gravity="end"  
    android:layout_marginTop="55dp"  
    android:layout_marginEnd="25dp"  
    app:srcCompat="@drawable/title2" />
```

```
<ImageView
```

```
    android:id="@+id/imageView5"  
    android:layout_width="140dp"  
    android:layout_height="45dp"  
    android:layout_marginStart="28dp"  
    android:layout_marginTop="90dp"  
    app:srcCompat="@drawable/title3" />
```

```
<LinearLayout
```

```
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:background="#9FFFFFFF"
android:orientation="vertical"
app:cardBackgroundColor="#ABFFFFFF">
```

```
<androidx.cardview.widget.CardView
    android:id="@+id/imageView"
    android:layout_width="match_parent"
    android:layout_height="60dp"
    android:layout_marginTop="150dp"
    app:cardElevation="0dp"
    android:background="#00FFFFFF"
    android:backgroundTint="#00FFFFFF">
```

```
<ImageView
    android:id="@+id/imageView2"
    android:layout_width="match_parent"
    android:layout_height="55dp"
    android:layout_marginStart="20dp"
    android:layout_marginEnd="20dp"
    android:scaleType="fitXY"
    app:srcCompat="@drawable/sousuo1" />
```

```
<LinearLayout
    android:id="@+id/findstudent"
    android:layout_width="match_parent"
```

```
android:layout_height="match_parent"
android:layout_marginStart="20dp"
android:layout_marginEnd="20dp"
android:orientation="horizontal">
```

```
<ImageButton
```

```
    android:id="@+id/findIcon"
    android:layout_width="35dp"
    android:layout_height="35dp"
    android:layout_marginStart="20dp"
    android:layout_marginTop="12dp"
    android:background="#00FFFFFF"
    android:contentDescription="TODO"
    android:scaleType="fitCenter"
    android:src="@drawable/sousuo_2" />
```

```
<EditText
```

```
    android:id="@+id/findInput"
    android:layout_width="309dp"
    android:layout_height="50dp"
    android:layout_marginStart="15dp"
    android:layout_marginTop="3dp"
    android:layout_marginEnd="10dp"
    android:layout_weight="1"
    android:hint="Input"
    android:textColor="@color/dark_black"
    android:textSize="26sp" />
```

```
</LinearLayout>
```

```
</androidx.cardview.widget.CardView>
```

```
<FrameLayout
```

```
    android:id="@+id/FrameLayout"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="635dp">
```

```
<LinearLayout
```

```
    android:id="@+id/more"
```

```
    android:layout_marginTop="45dp"
```

```
    android:layout_marginStart="0dp"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="500dp"
```

```
    android:orientation="vertical">
```

```
<LinearLayout
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="50dp"
```

```
    android:layout_gravity="bottom"
```

```
    android:layout_marginStart="10dp"
```

```
    android:layout_marginTop="10dp"
```

```
    android:orientation="horizontal">
```

```
<ImageView
```

```
    android:id="@+id/number"
```

```
    android:layout_width="40dp"
```

```
    android:layout_height="40dp"
```

```
        android:layout_marginStart="40dp"
        android:contentDescription="@string/todo"
        app:srcCompat="@drawable/biaoqian" />
```

```
<EditText
```

```
    android:id="@+id/number2"
    android:layout_width="260dp"
    android:layout_height="48dp"
    android:layout_marginStart="8dp"
    android:layout_marginEnd="45dp"
    android:ems="10"
    android:hint="@string/number"
    android:inputType="text" />
```

```
</LinearLayout>
```

```
<LinearLayout
```

```
    android:layout_width="match_parent"
    android:layout_height="50dp"
    android:layout_marginStart="10dp"
    android:layout_marginTop="20dp"
    android:orientation="horizontal">
```

```
<ImageView
```

```
    android:id="@+id/name"
    android:layout_width="40dp"
    android:layout_height="40dp"
    android:layout_marginStart="40dp"
    android:contentDescription="@string/todo"
    app:srcCompat="@drawable/people" />
```

```
<EditText
    android:id="@+id/name2"
    android:layout_width="260dp"
    android:layout_height="45dp"
    android:layout_marginStart="8dp"
    android:layout_marginEnd="45dp"
    android:ems="10"
    android:hint="name"
    android:inputType="text" />
</LinearLayout>
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="50dp"
    android:layout_marginStart="10dp"
    android:layout_marginTop="20dp"
    android:orientation="horizontal">
```

```
<ImageView
    android:id="@+id/email"
    android:layout_width="40dp"
    android:layout_height="40dp"
    android:layout_marginStart="40dp"
    android:contentDescription="@string/todo"
    app:srcCompat="@drawable/email" />
```

```
<EditText
    android:id="@+id/email2"
```

```
        android:layout_width="260dp"
        android:layout_height="45dp"
        android:layout_marginStart="8dp"
        android:layout_marginEnd="45dp"
        android:ems="10"
        android:hint="email"
        android:inputType="text" />
    </LinearLayout>
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="50dp"
    android:layout_marginStart="10dp"
    android:layout_marginTop="20dp"
    android:orientation="horizontal">
```

```
<ImageView
    android:id="@+id/location"
    android:layout_width="40dp"
    android:layout_height="40dp"
    android:layout_marginStart="40dp"
    android:contentDescription="@string/todo"
    app:srcCompat="@drawable/ditu" />
```

```
<EditText
    android:id="@+id/location2"
    android:layout_width="260dp"
    android:layout_height="45dp"
    android:layout_marginStart="8dp"
```

```

        android:layout_marginEnd="45dp"
        android:ems="10"
        android:hint="location"
        android:inputType="text" />
    </LinearLayout>
</LinearLayout>

<ListView
    android:id="@+id/UserList"
    android:layout_width="match_parent"
    android:layout_height="590dp"
    android:layout_margin="20dp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />

<ImageButton
    android:id="@+id/addButton"
    android:layout_width="80dp"
    android:layout_height="80dp"
    android:layout_gravity="bottom"
    android:layout_marginStart="300dp"
    android:layout_marginBottom="20dp"
    android:background="#00FFFFFF"
    android:contentDescription="TODO"
    android:scaleType="fitCenter"
    app:srcCompat="@drawable/add" />
</FrameLayout>

```

```
<androidx.coordinatorlayout.widget.CoordinatorLayout
    android:layout_width="match_parent"
    android:layout_height="80dp"
    android:layout_marginTop="0dp"
    android:background="@color/yellowLight">
```

```
<com.google.android.material.bottomnavigation.BottomNavigationView
    android:id="@+id/bottomNavigationView"
    android:layout_width="match_parent"
    android:layout_height="80dp"
    android:layout_gravity="bottom"
    android:background="@color/yellowLight">
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_gravity="center_vertical"
    android:baselineAligned="false">
```

```
<ImageButton
    android:id="@+id/user"
    android:layout_width="80dp"
    android:layout_height="60dp"
    android:layout_weight="3"
    android:background="#00FFFFFF"
    android:contentDescription="TODO"
    android:scaleType="fitCenter"
    android:src="@drawable/tuceng28" />
```

```
<ImageButton
    android:id="@+id/all"
    android:layout_width="80dp"
    android:layout_height="60dp"
    android:layout_weight="3"
    android:background="#00FFFFFF"
    android:contentDescription="TODO"
    android:scaleType="fitCenter"
    android:src="@drawable/tuceng12" />
```

```
<ImageButton
    android:id="@+id/setting"
    android:layout_width="80dp"
    android:layout_height="60dp"
    android:layout_weight="3"
    android:background="#00FFFFFF"
    android:contentDescription="TODO"
    android:scaleType="fitCenter"
    android:src="@drawable/tuceng27" />
```

```
</LinearLayout>
```

```
</com.google.android.material.bottomnavigation.BottomNavigationView>
```

```
</androidx.coordinatorlayout.widget.CoordinatorLayout>
```

```
</LinearLayout>
```

```
<LinearLayout
```

```
    android:id="@+id/edit"
```

```
android:layout_width="match_parent"
android:layout_height="match_parent"
android:background="#9FFFFFFF"
android:orientation="vertical"
app:cardBackgroundColor="#ABFFFFFF">
```

```
<com.google.android.material.bottomnavigation.BottomNavigationView
```

```
    android:layout_width="match_parent"
    android:layout_height="50dp"
    android:layout_gravity="top"
    app:elevation="0dp"
    android:layout_marginTop="150dp"
    android:layout_margin="20dp"
    android:background="#00FFFFFF">
```

```
<LinearLayout
```

```
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="horizontal">
```

```
<ImageButton
```

```
    android:id="@+id/exit"
    android:layout_width="50dp"
    android:layout_height="50dp"
    android:background="#00FFFFFF"
    android:contentDescription="TODO"
    android:scaleType="fitCenter"
    app:srcCompat="@drawable/last" />
```

```
<ImageButton
    android:id="@+id/addok"
    android:layout_width="50dp"
    android:layout_height="50dp"
    android:layout_marginStart="260dp"
    android:background="#00FFFFFF"
    android:contentDescription="TODO"
    android:scaleType="fitCenter"
    app:srcCompat="@drawable/dui1" />
</LinearLayout>
```

```
</com.google.android.material.bottomnavigation.BottomNavigationView>
```

```
<TextView
    android:id="@+id/ID_input2"
    android:layout_width="322dp"
    android:layout_height="wrap_content"
    android:layout_marginStart="110dp"
    android:layout_marginTop="100dp"
    android:text="@string/请输入您的学号"
    android:textColor="@color/dark"
    android:textStyle="italic" />
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="50dp"
    android:layout_gravity="bottom"
```

```
android:layout_marginStart="10dp"
android:layout_marginTop="10dp"
android:orientation="horizontal">
```

```
<ImageView
```

```
    android:id="@+id/peopleIDinput"
    android:layout_width="50dp"
    android:layout_height="50dp"
    android:layout_marginStart="40dp"
    android:contentDescription="@string/todo"
    app:srcCompat="@drawable/biaoqian" />
```

```
<EditText
```

```
    android:id="@+id/IDinput"
    android:layout_width="260dp"
    android:layout_height="50dp"
    android:layout_marginStart="8dp"
    android:layout_marginEnd="45dp"
    android:ems="10"
    android:hint="number"
    android:inputType="text" />
```

```
</LinearLayout>
```

```
<TextView
```

```
    android:id="@+id/PASSWORD_input2"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginStart="110dp"
    android:layout_marginTop="15dp"
```

```
android:text="@string/请输入您的姓名"  
android:textColor="@color/dark"  
android:textStyle="italic" />
```

<LinearLayout

```
android:layout_width="match_parent"  
android:layout_height="50dp"  
android:layout_marginStart="10dp"  
android:layout_marginTop="10dp"  
android:orientation="horizontal">
```

<ImageView

```
android:id="@+id/peopleNameinput"  
android:layout_width="50dp"  
android:layout_height="50dp"  
android:layout_marginStart="40dp"  
android:contentDescription="@string/todo"  
app:srcCompat="@drawable/people" />
```

<EditText

```
android:id="@+id/Nameinput"  
android:layout_width="260dp"  
android:layout_height="50dp"  
android:layout_marginStart="8dp"  
android:layout_marginEnd="45dp"  
android:ems="10"  
android:hint="name"  
android:inputType="text" />
```

</LinearLayout>

<TextView

```
    android:id="@+id/Email"
    android:layout_width="322dp"
    android:layout_height="wrap_content"
    android:layout_marginStart="110dp"
    android:layout_marginTop="15dp"
    android:text="请输入您的邮箱"
    android:textColor="@color/dark"
    android:textStyle="italic" />
```

<LinearLayout

```
    android:layout_width="match_parent"
    android:layout_height="50dp"
    android:layout_gravity="bottom"
    android:layout_marginStart="10dp"
    android:layout_marginTop="10dp"
    android:orientation="horizontal">
```

<ImageView

```
    android:id="@+id/EmailInput"
    android:layout_width="50dp"
    android:layout_height="50dp"
    android:layout_marginStart="40dp"
    android:contentDescription="@string/todo"
    app:srcCompat="@drawable/email" />
```

<EditText

```
    android:id="@+id/EMAILInput"
```

```

        android:layout_width="260dp"
        android:layout_height="50dp"
        android:layout_marginStart="8dp"
        android:layout_marginEnd="45dp"
        android:ems="10"
        android:hint="email"
        android:inputType="text" />
    </LinearLayout>

    <TextView
        android:id="@+id/location_input2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginStart="110dp"
        android:layout_marginTop="15dp"
        android:text="@string/请输入您所在的省份"
        android:textColor="@color/dark"
        android:textStyle="italic" />

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="50dp"
        android:layout_marginStart="10dp"
        android:layout_marginTop="10dp"
        android:orientation="horizontal">

        <ImageView
            android:id="@+id/LOCATIONinput"
            android:layout_width="50dp"

```

<pre> android:layout_height="50dp" android:layout_marginStart="40dp" android:contentDescription="@string/todo" app:srcCompat="@drawable/ditu" /> <EditText android:id="@+id/Locationinput" android:layout_width="260dp" android:layout_height="50dp" android:layout_marginStart="8dp" android:layout_marginEnd="45dp" android:ems="10" android:hint="location" android:inputType="text" /> </LinearLayout> <!-- <FrameLayout--> <!-- android:layout_width="match_parent"--> <!-- android:layout_height="match_parent">--> <!-- </FrameLayout>--> </LinearLayout> </FrameLayout> </pre>
主要源代码

```

# MainActivity.java

@Override

protected void onCreate(Bundle savedInstanceState) {

    super.onCreate(savedInstanceState);

    setContentView(R.layout.activity_main);

    EdgeToEdge.enable(this);

ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.backgroun
d), (v, insets) -> {

        Insets                systemBars                =
insets.getInsets(WindowInsetsCompat.Type.systemBars());

        v.setPadding(systemBars.left,                systemBars.top,
systemBars.right, systemBars.bottom);

        return insets;

    });

    // 设置全屏标志

//

getWindow().setFlags(WindowManager.LayoutParams.FLAG_FULLSCREEN,
//
                WindowManager.LayoutParams.FLAG_FULLSCREEN);

getWindow().setSoftInputMode(WindowManager.LayoutParams.SOFT_INPUT_ST
ATE_HIDDEN);

getWindow().setFlags(WindowManager.LayoutParams.FLAG_FULLSCREEN,
WindowManager.LayoutParams.FLAG_FULLSCREEN);

    View decorView = getWindow().getDecorView();

```

```

        int uiOptions = View.SYSTEM_UI_FLAG_HIDE_NAVIGATION |
View.SYSTEM_UI_FLAG_IMMERSIVE_STICKY;

        decorView.setSystemUiVisibility(uiOptions);

        LinearLayout loginLayout = findViewById(R.id.login);
        loginLayout.setVisibility(View.VISIBLE);

        FrameLayout appLayout = findViewById(R.id.app);
        appLayout.setVisibility(View.GONE);

        LinearLayout mainLayout = findViewById(R.id.main);
        mainLayout.setVisibility(View.GONE);

        LinearLayout moreLayout = findViewById(R.id.more);
        moreLayout.setVisibility(View.GONE);

        LinearLayout editLayout = findViewById(R.id.edit);
        editLayout.setVisibility(View.GONE);

        findViewById(R.id.FrameLayout);

        helper=new StudentDatabaseHelper( MainActivity.this);

        Button loginBTN = findViewById(R.id.login_btn);
        loginBTN.setOnClickListener(v -> {

            EditText idInput = findViewById(R.id.ID);
            EditText passwordInput = findViewById(R.id.password);

            String inputId = idInput.getText().toString().trim();

```

```

        String                inputPassword                =
passwordInput.getText().toString().trim();

        if (inputId.isEmpty() || inputPassword.isEmpty()) {
            Toast.makeText(MainActivity.this, "请输入学号和密码",
Toast.LENGTH_SHORT).show();

            return;
        }

        try (SQLiteDatabase db = helper.getWritableDatabase()) {
            ContentValues values = new ContentValues();
            values.put("STUDENTID", inputId); // ✓ 改对字段
            values.put("PASSWORD", inputPassword);

            long newRowId = db.insert("USER", null, values);

            if (newRowId != -1) {
                Toast.makeText(MainActivity.this, "注册成功!",
Toast.LENGTH_SHORT).show();

                loginLayout.setVisibility(View.GONE);
                appLayout.setVisibility(View.VISIBLE);
                mainLayout.setVisibility(View.VISIBLE);
            } else {
                Toast.makeText(MainActivity.this, "注册失败, 该用
户名已存在!", Toast.LENGTH_SHORT).show();
            }
        } catch (Exception e) {
            e.printStackTrace();

            Toast.makeText(MainActivity.this, "注册异常!",

```

```

Toast.LENGTH_SHORT).show();

    }

});

Button registerBTN = findViewById(R.id.register_btn);
registerBTN.setOnClickListener(new View.OnClickListener() {

    @Override

    public void onClick(View v) {

        EditText idInput = findViewById(R.id.ID);

        EditText passwordInput = findViewById(R.id.password);

        String inputId = idInput.getText().toString().trim();

        String                inputPassword                =
passwordInput.getText().toString().trim();

        if (inputId.isEmpty() || inputPassword.isEmpty()) {

            Toast.makeText(MainActivity.this, "请输入学号和密
码", Toast.LENGTH_SHORT).show();

            return;

        }

        SQLiteDatabase db = helper.getReadableDatabase();

        Cursor cursor = db.rawQuery(

            "SELECT * FROM USER WHERE STUDENTID = ? AND
PASSWORD = ?",

            new String[] {inputId, inputPassword}

        );

```

```

        if (cursor.moveToFirst()) {
            Toast.makeText(MainActivity.this, "登录成功!",
Toast.LENGTH_SHORT).show();

            loginLayout.setVisibility(View.GONE);
            appLayout.setVisibility(View.VISIBLE);
            mainLayout.setVisibility(View.VISIBLE);
        } else {
            Toast.makeText(MainActivity.this, "账号或密码错
误!", Toast.LENGTH_SHORT).show();
        }

        cursor.close();
        db.close();
    }
});

// ===== 显示学号列表 =====
ListView userList = findViewById(R.id.UserList);
StudentDatabaseHelper dbHelper = new
StudentDatabaseHelper(this);
SQLiteDatabase db = dbHelper.getReadableDatabase();

Cursor cursor = db.rawQuery("SELECT number FROM student", null);

ArrayList<String> numberList = new ArrayList<>();
if (cursor.moveToFirst()) {
    do {
        String number =
cursor.getString(cursor.getColumnIndexOrThrow("number"));

```

```

        numberList.add(number);
    } while (cursor.moveToNext());
}

cursor.close();
db.close();

// 用自定义 list_item_main 布局
ArrayAdapter<String> adapter = new ArrayAdapter<>(this,
R.layout.list_item_main, R.id.textViewItem, numberList);
userList.setAdapter(adapter);

// ===== 详细信息 =====
// 找到按钮
ImageButton addButton = findViewById(R.id.addButton);
ImageButton addok = findViewById(R.id.addok);

userList.setOnItemClickListener((parent, view, position, id) ->
{
    // 1. 获取点击的 number
    String selectedNumber = numberList.get(position);

    // 2. 查询数据库, 拿到该 number 对应的完整信息
    SQLiteDatabase db1 = helper.getReadableDatabase();
    Cursor cursor1 = db1.rawQuery("SELECT * FROM student WHERE
number = ?", new String[] {selectedNumber});

    if (cursor1.moveToFirst()) {
        String number =
        cursor1.getString(cursor1.getColumnIndexOrThrow("number"));
    }
}

```

```

        String                name                =
cursor1.getString(cursor1.getColumnIndexOrThrow("name"));

        String                email                =
cursor1.getString(cursor1.getColumnIndexOrThrow("email"));

        String                location                =
cursor1.getString(cursor1.getColumnIndexOrThrow("location"));

// 3. 显示到 more 布局

EditText number2 = findViewById(R.id.number2);
EditText name2 = findViewById(R.id.name2);
EditText email2 = findViewById(R.id.email2);
EditText location2 = findViewById(R.id.location2);

number2.setText(number);
name2.setText(name);
email2.setText(email);
location2.setText(location);

// 4. 显示 more, 隐藏其他可点击, main 禁用
userList.setVisibility(View.GONE);
moreLayout.setVisibility(View.VISIBLE);
addButton.setVisibility(View.GONE);

appLayout.setEnabled(false);
mainLayout.setEnabled(false);
loginLayout.setEnabled(false);
editLayout.setEnabled(false);

// 可以顺便给背景加一层透明蒙版（美观效果）

```

```

    }

    cursor1.close();
    db1.close();
});

// 5. 给 more 布局设置点击事件, 点击返回 main 布局
moreLayout.setOnClickListener(v -> {
    moreLayout.setVisibility(View.GONE);
    userList.setVisibility(View.VISIBLE);
    addButton.setVisibility(View.VISIBLE);

    // 恢复其他布局的可点击
    appLayout.setEnabled(true);
    mainLayout.setEnabled(true);
    loginLayout.setEnabled(true);
    editLayout.setEnabled(true);
});

// ===== 删除操作 =====

// 给 UserList 的 item 添加长按删除功能
userList.setOnItemLongClickListener((parent, view, position,
id) -> {
    String selectedNumber = numberList.get(position); // 拿到
被长按的 number

    // 弹出确认对话框
    new AlertDialog.Builder(MainActivity.this)

```

```

        .setTitle("删除确认")
        .setMessage("确定要删除学号为 " + selectedNumber +
" 的学生吗? ")
        .setPositiveButton("确定", (dialog, which) -> {
            try {

                // 执行删除操作, 注意条件写成 "number = ?"
并传参数

                int rowsDeleted = db.delete("STUDENT",
"number = ?", new String[] {selectedNumber});

                if (rowsDeleted > 0) {
                    Toast.makeText(MainActivity.this, "删
除成功!", Toast.LENGTH_SHORT).show();

                    // 同时从列表里删除该项
                    numberList.remove(position);
                    adapter.notifyDataSetChanged();
                } else {
                    Toast.makeText(MainActivity.this, "删
除失败, 找不到该学号!", Toast.LENGTH_SHORT).show();
                }

            } catch (Exception e) {
                Log.e("DELETE_ERROR", "删除操作失败: " +
e.getMessage(), e);
                Toast.makeText(MainActivity.this, "删除失
败, 操作异常!", Toast.LENGTH_SHORT).show();
            } finally {

```

```

        if (db != null && db.isOpen()) {
            db.close();
        }
    })
    .setNegativeButton("取消", null)
    .show();

return true; // 注意：长按事件必须返回 true，否则还会触发
短按

});

// ===== 查找功能 =====

// 找到输入框和查询按钮
EditText findInput = findViewById(R.id.findInput);
ImageButton findIcon = findViewById(R.id.findIcon);

// 给 findIcon 添加点击事件
findIcon.setOnClickListener(v -> {
    String searchText = findInput.getText().toString().trim();

    SQLiteDatabase db1 = helper.getReadableDatabase();
    Cursor searchCursor;

    if (searchText.isEmpty()) {
        // 输入为空，查询所有
        searchCursor = db1.rawQuery("SELECT number FROM

```

```

student", null);

        } else {

            // 输入不为空，进行模糊查询（支持中文）

            searchCursor = db1.rawQuery(

                "SELECT number FROM student WHERE number LIKE ?

OR name LIKE ? OR email LIKE ? OR location LIKE ?",

                new String[] {"%" + searchText + "%", "%" +

searchText + "%", "%" + searchText + "%", "%" + searchText + "%"}

            );

        }

        ArrayList<String> searchResultList = new ArrayList<>();

        if (searchCursor.moveToFirst()) {

            do {

                String number =

searchCursor.getString(searchCursor.getColumnIndexOrThrow("number"));

                searchResultList.add(number);

            } while (searchCursor.moveToNext());

        }

        searchCursor.close();

        db1.close();

        // 更新 ListView

        ArrayAdapter<String> searchAdapter = new

ArrayAdapter<>(MainActivity.this, R.layout.list_item_main,

R.id.textViewItem, searchResultList);

        userList.setAdapter(searchAdapter);

        if (searchResultList.isEmpty()) {

```

```

        Toast.makeText(MainActivity.this, "未找到相关记录",
Toast.LENGTH_SHORT).show();

    }

});

// 找到输入框
EditText numberInput = findViewById(R.id.IDinput);
EditText nameInput = findViewById(R.id.Nameinput);
EditText emailInput = findViewById(R.id.EMAILinput);
EditText locationInput = findViewById(R.id.Locationinput);

// 点击 addButton, 显示 edit 界面
addButton.setOnClickListener(v -> {
    mainLayout.setVisibility(View.GONE);
    editLayout.setVisibility(View.VISIBLE);
});

// 点击 addokButton, 保存到数据库
addok.setOnClickListener(v -> {
    String number = numberInput.getText().toString().trim();
    String name = nameInput.getText().toString().trim();
    String email = emailInput.getText().toString().trim();
    String          location          =
locationInput.getText().toString().trim();

    if (number.isEmpty() || name.isEmpty() || email.isEmpty() ||
location.isEmpty()) {
        Toast.makeText(MainActivity.this, "请完整填写所有信息",
Toast.LENGTH_SHORT).show();
    }
}
});

```

```

        return;
    }

    SQLiteDatabase db2 = helper.getWritableDatabase();

    // 先查询 number 是否已存在
    Cursor cursorCheck = db2.rawQuery(
        "SELECT * FROM student WHERE number = ?", new
String[] {number}
    );

    if (cursorCheck.moveToFirst()) {
        // number 已存在, 提示错误
        Toast.makeText(MainActivity.this, "学号已存在, 请重新
输入", Toast.LENGTH_SHORT).show();
    } else {
        // 插入新数据
        ContentValues values = new ContentValues();
        values.put("number", number);
        values.put("name", name);
        values.put("email", email);
        values.put("location", location);

        long insertResult = db2.insert("student", null,
values);

        if (insertResult != -1) {
            Toast.makeText(MainActivity.this, "添加成功!",
Toast.LENGTH_SHORT).show();

```

```

        // 切换回 mainLayout, 刷新列表
        editLayout.setVisibility(View.GONE);
        mainLayout.setVisibility(View.VISIBLE);

        refreshUserList(); //    重新加载列表
    } else {
        Toast.makeText(MainActivity.this, "添加失败!",
Toast.LENGTH_SHORT).show();
    }
}

    cursorCheck.close();
    db2.close();
});

}

// 刷新列表的方法
private void refreshUserList() {
    ListView userList = findViewById(R.id.UserList);
    SQLiteDatabase db = helper.getReadableDatabase();
    Cursor cursor = db.rawQuery("SELECT number FROM student", null);

    ArrayList<String> numberList = new ArrayList<>();
    if (cursor.moveToFirst()) {
        do {
            String                number                =
cursor.getString(cursor.getColumnIndexOrThrow("number"));

```

```

        numberList.add(number);
    } while (cursor.moveToNext());
}

cursor.close();

db.close();

ArrayAdapter<String> adapter = new ArrayAdapter<>(this,
R.layout.list_item_main, R.id.textViewItem, numberList);

userList.setAdapter(adapter);
}
}

# StudentDatabaseHelper.java
public class StudentDatabaseHelper extends SQLiteOpenHelper {
    // 数据库名称
    private static final String DBNAME = "student.db";
    // 数据库版本
    private static final int VERSION = 1;
    private Context mContext;
    // 日志插入结果
    private static void insertUser(SQLiteDatabase db, String
studentid, String password) {
        ContentValues userValues = new ContentValues();
        userValues.put("STUDENTID", studentid);
        userValues.put("PASSWORD", password);
        long result = db.insert("USER", null, userValues);
        Log.d("sqlite", "insert" + studentid + ",_id:" + result);
    }
    private static void insertStudent(SQLiteDatabase db, String name,
String number, String email, String location) {
        ContentValues studentValues = new ContentValues();

```

```

        studentValues.put("name", name);
        studentValues.put("number", number);
        studentValues.put("email", email);
        studentValues.put("location", location);

        long result = db.insert("STUDENT", null, studentValues);
        Log.d("sqlite", "insert student: " + name + ", rowid: " + result);
    }

    public StudentDatabaseHelper(@Nullable Context context) {
        super(context, DBNAME, null, VERSION);
        mContext = context;
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        // 创建 USER 表
        db.execSQL("CREATE TABLE USER (_id INTEGER PRIMARY KEY
AUTOINCREMENT, " +
            "STUDENTID TEXT, " +
            "PASSWORD TEXT);");
    };

    // 插入用户信息
    insertUser(db, "admin", "123456");

    // 创建 STUDENT 表
    db.execSQL("CREATE TABLE STUDENT (" +
        "name TEXT, " +
        "number TEXT PRIMARY KEY, " +
        "email TEXT, " +
        "location TEXT" +
        ");");

```

```

    );
    // 插入三条学生信息
    insertStudent(db, " 陈 梓 欣 ", "223428010210",
"chenzixin@stu.sau.edu.cn", "辽宁");
    insertStudent(db, " 郑 羽 婷 ", "223428010209",
"zhengyuting@stu.sau.edu.cn", "辽宁");
    insertStudent(db, " 刘 渤 ", "223428010103",
"liubo@stu.sau.edu.cn", "辽宁");
}
@Override
public void onUpgrade(SQLiteDatabase db, int oldVersion, int
newVersion) {
    db.execSQL("drop table if exists USER");
    db.execSQL("drop table if exists STUDENT");
    onCreate(db);
}
}

```

实验截图

用户登陆界面，输入用户名和密码进行登录，如图 1 所示。

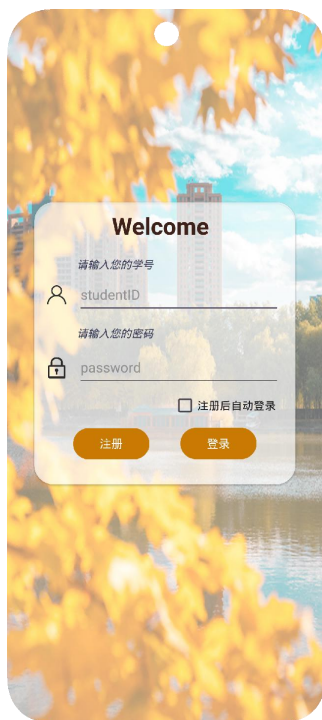


图 1 用户登陆界面

正确登录/注册后，进入 APP 主界面，如图 2 所示。

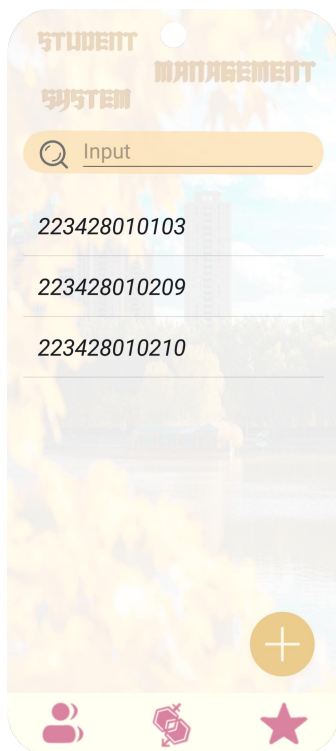


图 2 APP 主界面

增加学生信息，输入学号姓名和省份，进行添加操作，如图 3 所示。

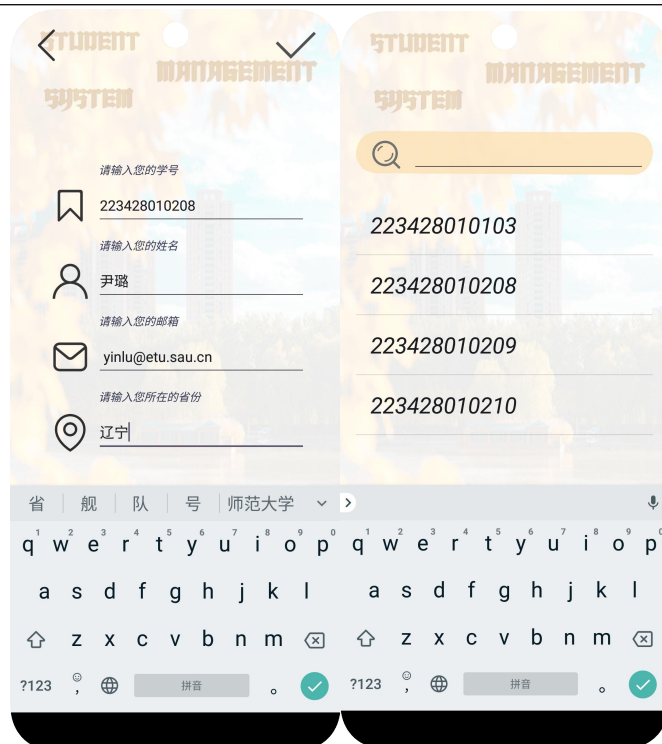


图 3 增加用户信息

输入学号、姓名、省份三者之一进行查询，如图 4 所示

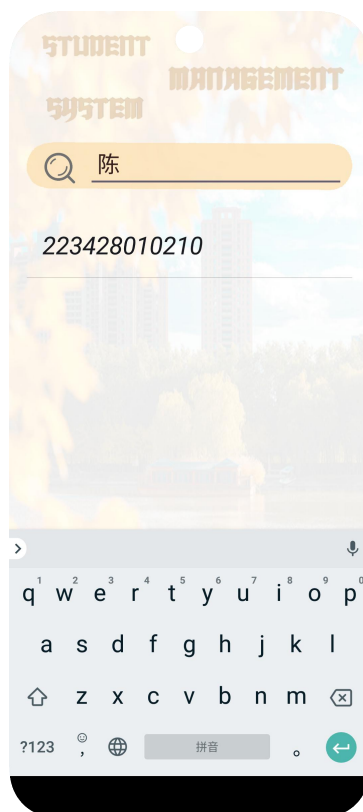


图 4 查询学生信息

在更改/删除界面进行删除更改操作，如图 5 所示。

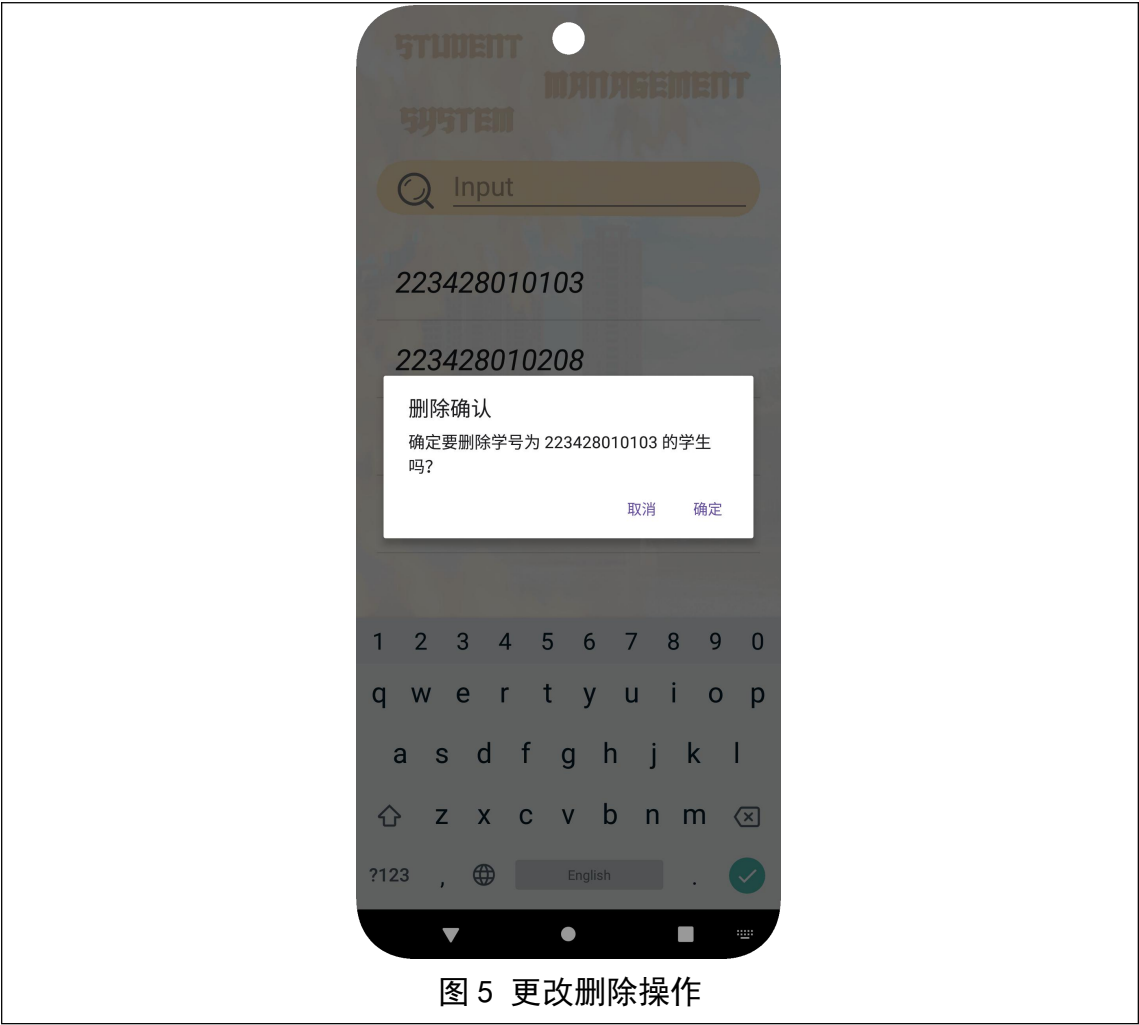


图 5 更改删除操作